Lab 2
Bipolar Junction Transistor (BJT)

Electronics and Photonics Post-undergraduate Second Degree Program

National Yang Ming Chiao Tung University

Outline

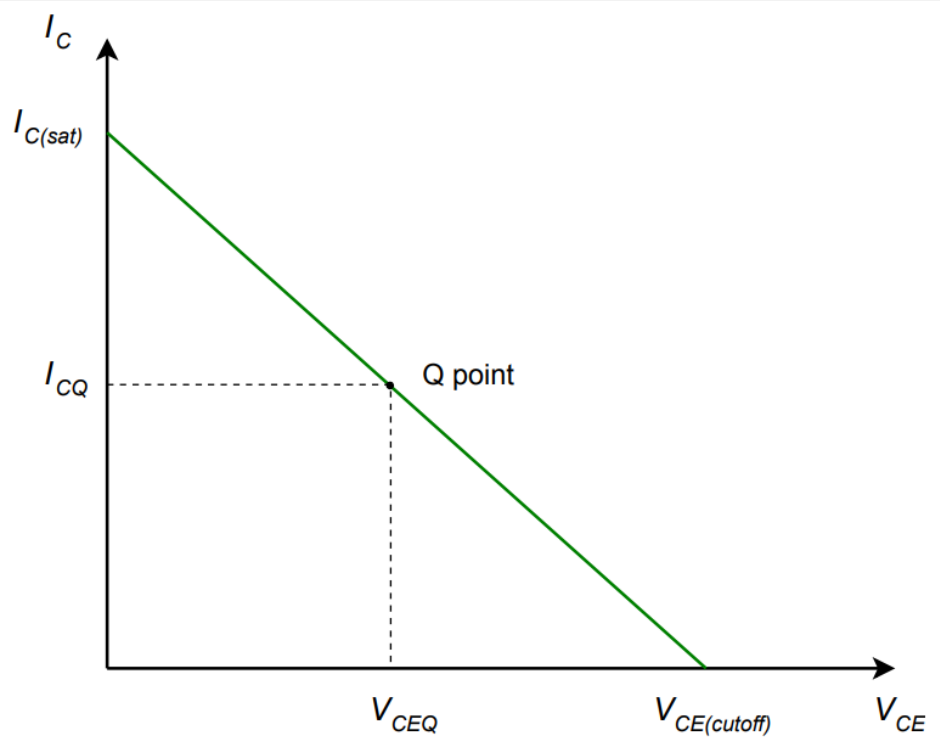
01 BJT Biasing & Characteristic Curve

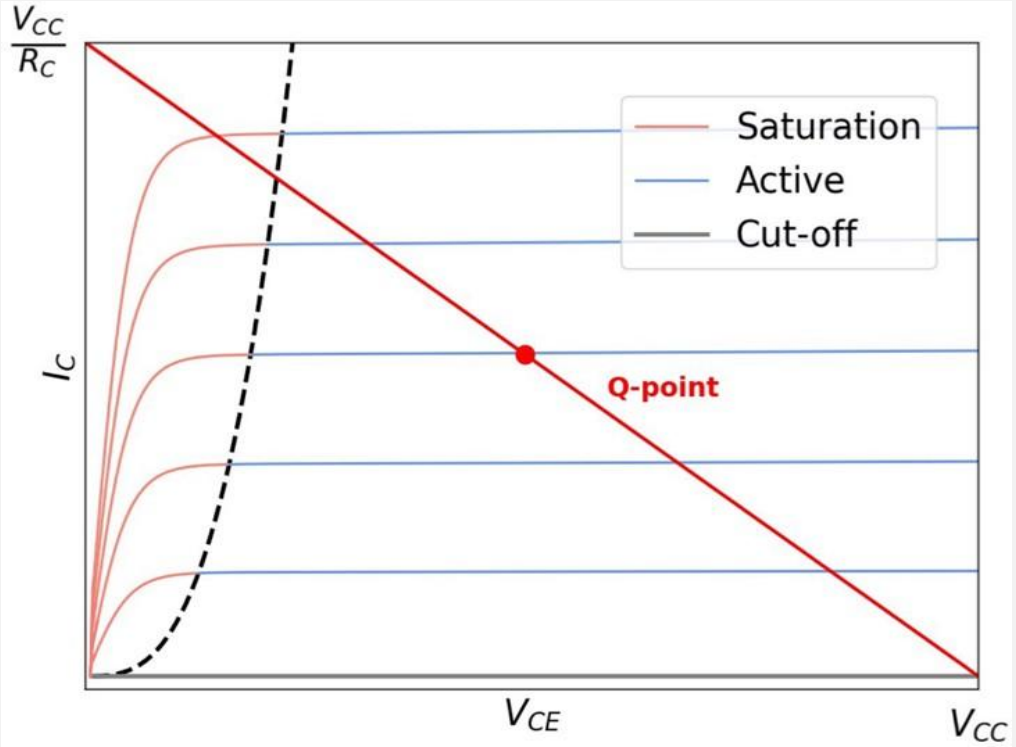
02 Common Emitter Amplifier (Voltage Gain)

BJT Biasing & Characteristic Curve

Objective

- Examine the voltage divider bias configuration to determine whether it can produce a stable quiescent operating point (Q-point).





Theory overview

Emitter (E) : Heavily N doped (n^{++})

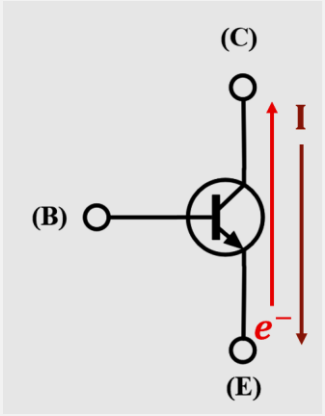
Function : Injects a large number of majority carriers (electrons in NPN).

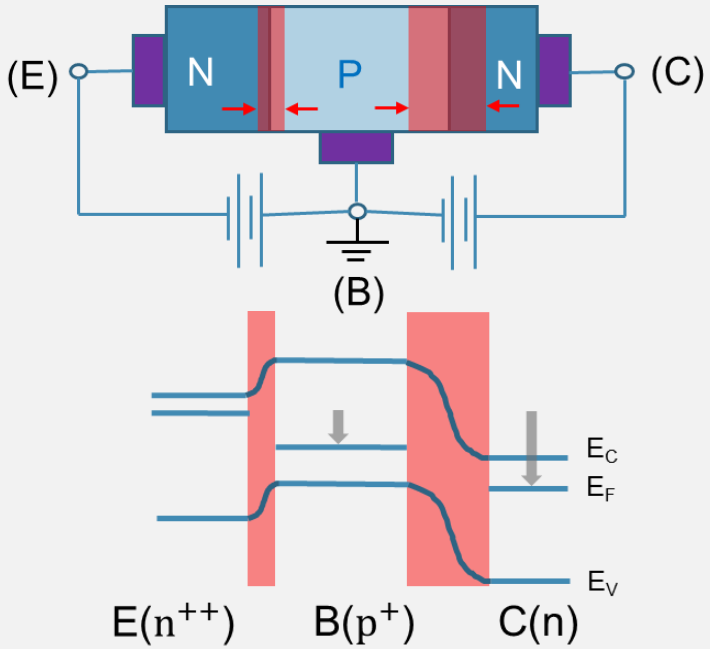
Base (B) : Lightly P doped (p^{+})

Function : Controls the number of carriers, but its current is very small.

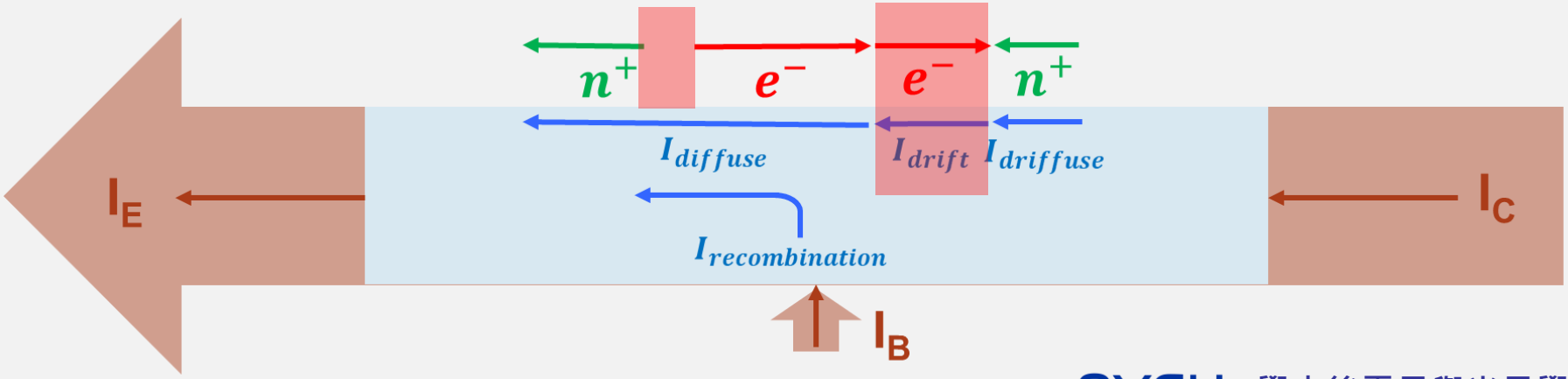
Collector (C) : Moderately N doped (n)

Function : collects most of the carriers injected from the emitter.





Forward bias Reverse bias



Theory overview

α : Common-Base Current Gain

$$\alpha = \frac{I_C}{I_E}$$

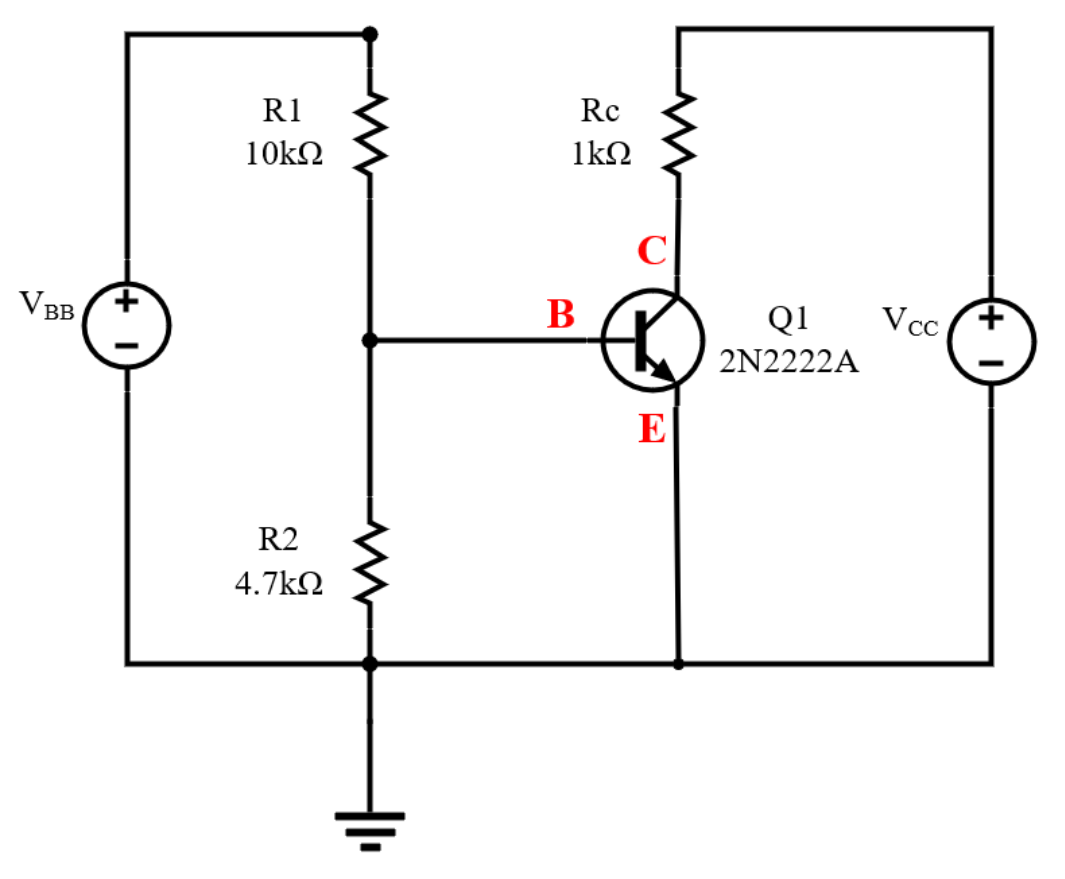
- Definition : The rate of Emitter current be collected by collector
- Range : 0.98 ~ 0.998, very close to 1 ; The other part is recombined as I_B in the base.

β : Common-Emitter Current Gain

$$\beta = \frac{I_C}{I_B}$$

- Definition : Represents how much the base current is amplified into the collector current.
- Range : 50 ~ 200

Procedure



Characteristic curve

1. Set up the circuits.

2. By adjusting different V_{BB}, the value of V_{BE} can be determined (measured with a multimeter).

3. After fixing V_{BE}, increase V_{CC} to determine V_{CE} (measured with a multimeter)

4. Measure the current I_C.

5. Plot the BJT characteristic curves.

(Table1.)

Procedure

- Consider the circuit of Fig 4. using V_{CC} , R_1 = 10 kΩ, R_2 = 4.7 kΩ, and R_C = 1 kΩ.
- Set different V_{BE} and V_{CE} values to measure the collector current I_C, and plot the BJT characteristic curves.

V_{BB} = 1.5 V 、V_{BE} = V 、ON/OFF	
V_{CE} (experimental setting)	I_C (measure) $(=\frac{V_C}{R_C})$
V	A
V	A
V	A
V	A
V	A

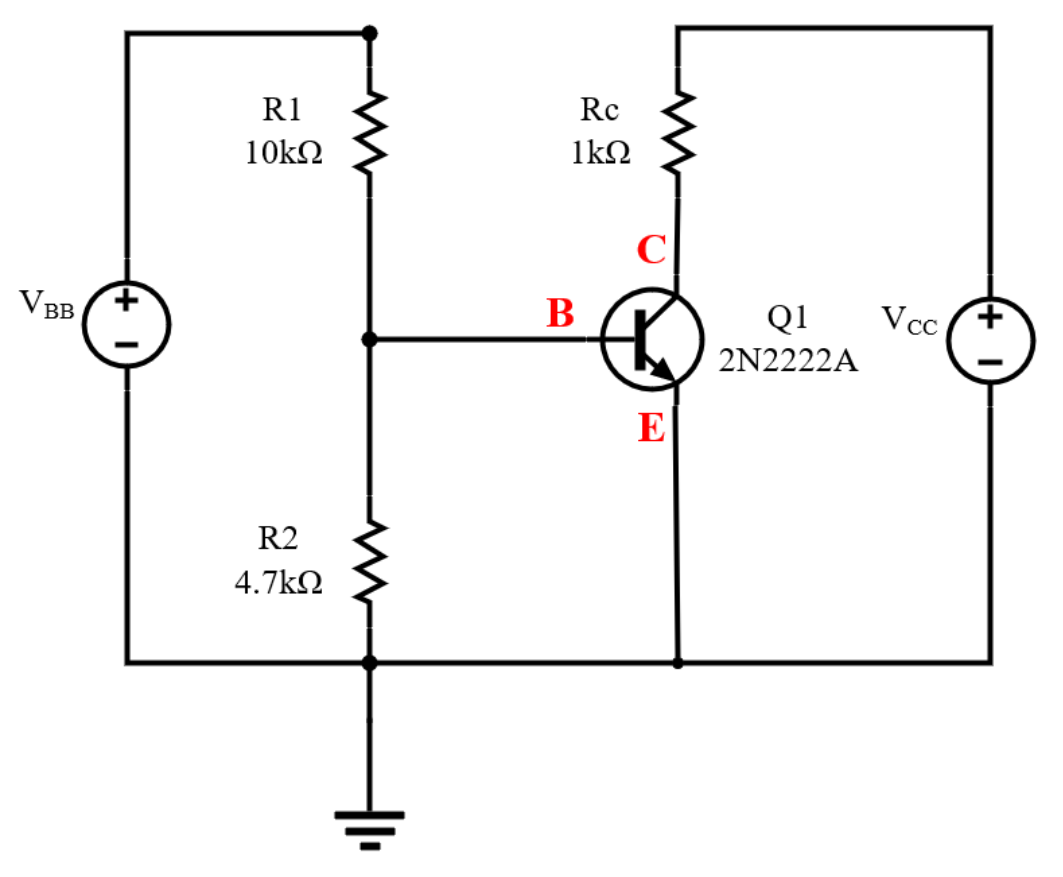
V_{BB} = 2 V 、V_{BE} = V 、ON/OFF	
V_{CE} (experimental setting)	I_C (measure) $(=\frac{V_C}{R_C})$
V	A
V	A
V	A
V	A
V	A

V_{BB} = 2.2 V 、V_{BE} = V 、ON/OFF	
V_{CE} (experimental setting)	I_C (measure) $(=\frac{V_C}{R_C})$
V	A
V	A
V	A
V	A
V	A

V_{BB} = 2.4 V 、V_{BE} = V 、ON/OFF	
I_C (measure) $(=\frac{V_C}{R_C})$	I_C (measure) $(=\frac{V_C}{R_C})$
V	A
V	A
V	A
V	A
V	A

Table 1

Procedure



1. Set up the circuits.

2. Measure the voltage difference across R1 and R2 respectively, and calculate the current values.

3. According to Kirchhoff's Current Law (KCL), the value of the base current I_B can be obtained.

4. Measure the voltage difference across R_C to determine the collector current I_C.

5. Calculate the β obtained from I_B and I_C, and compare it with the β measured using the multimeter.

(Table2.)

Procedure

- Record the results in the first row of Table 3. Based on these data, calculate and record the experimentally obtained β (current gain).

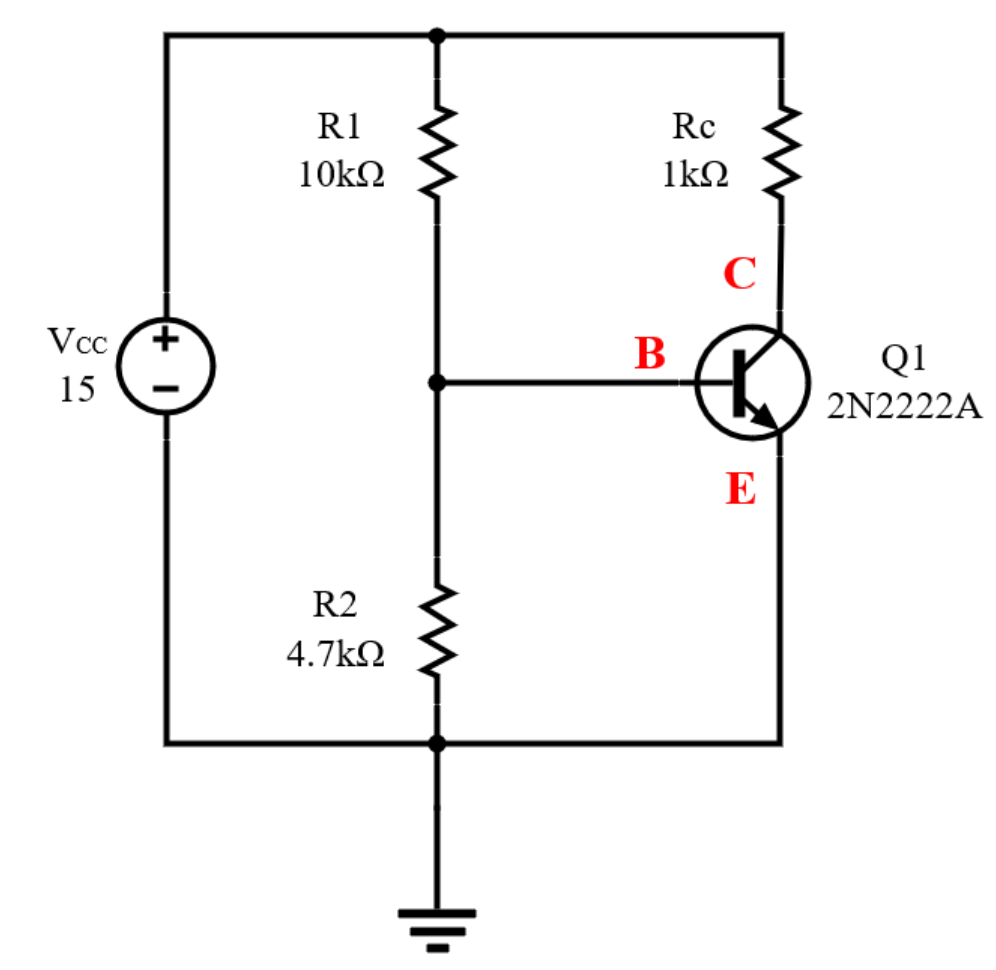
	I_B	I_C	$\beta_{Calculate}(I_C/I_B)$	β_{Meas}
BJT				

Table 2

Component List

Component	Specification	Number
BJT	2N2222A	1
Resistor	1 kΩ (1/4 W)	1
	4.7 kΩ (1/4 W)	1
	10 kΩ (1/4 W)	1
DC Power Supply	GW Instek GPE-3323	1
Signal Source (Function Generator)	GW Instek AFG-2225	1
Oscilloscope	Keysight EDUX1052G	1
Multimeter	GW Instek GDM-532	1

Supplement



Ideal endpoints of the DC load line and the quiescent operating point (Q-point).

Set up the circuits.

- For saturation, assume V_{CE} goes to 0

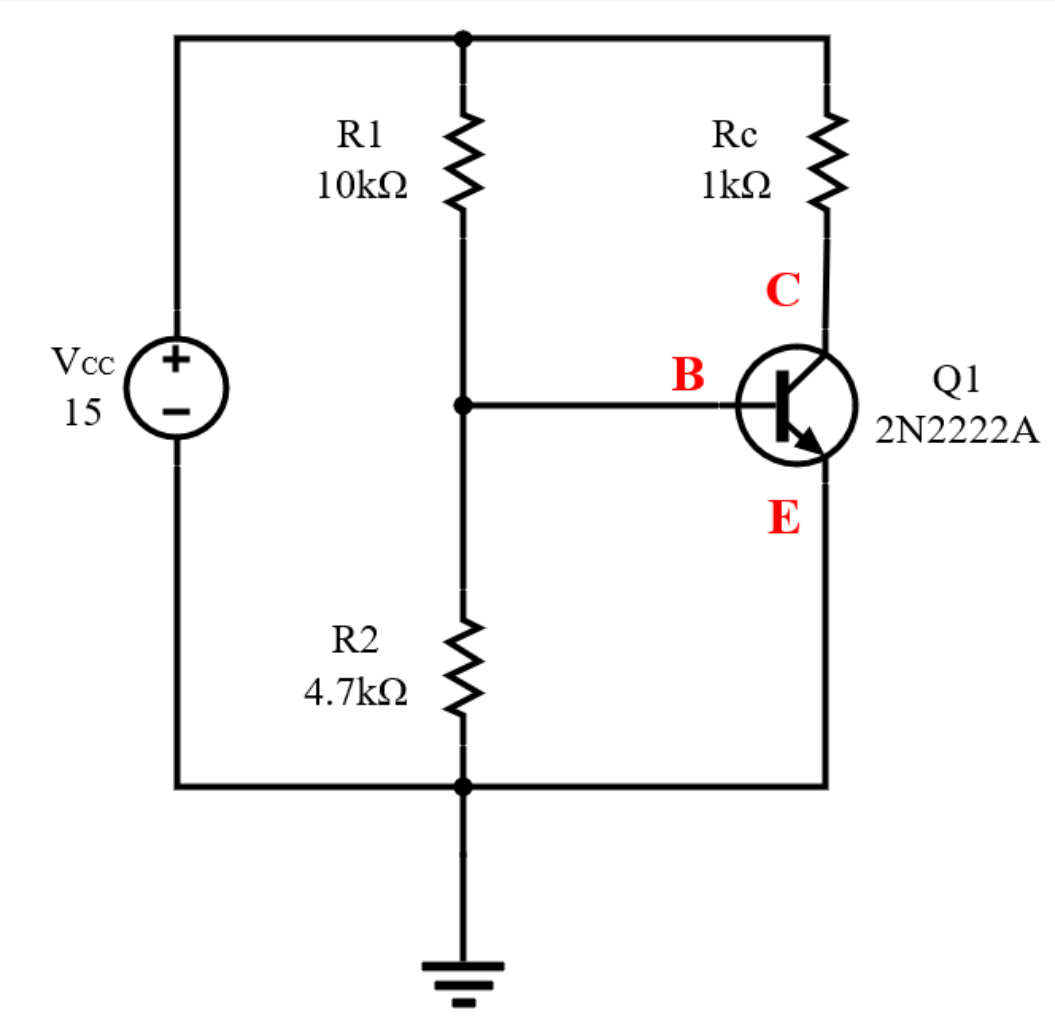
$$I_{C(sat)} = \frac{V_{CC}}{R_C}$$

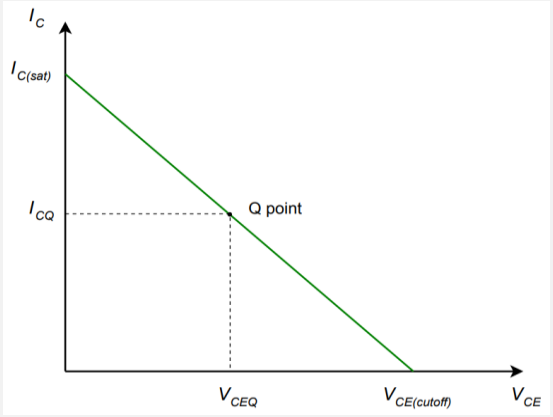
$V_{CE(cutoff)}$ occurs when $I_C = 0$

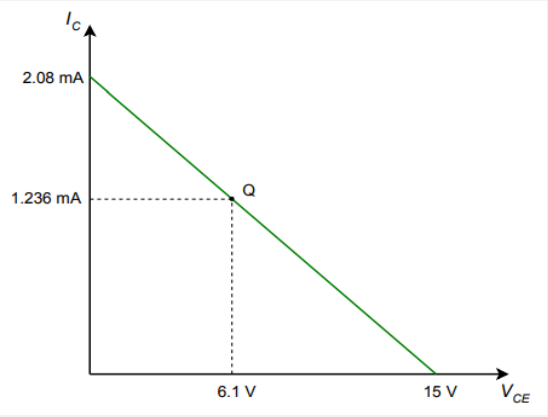
$$V_{CE(cutoff)} = V_{CC}$$

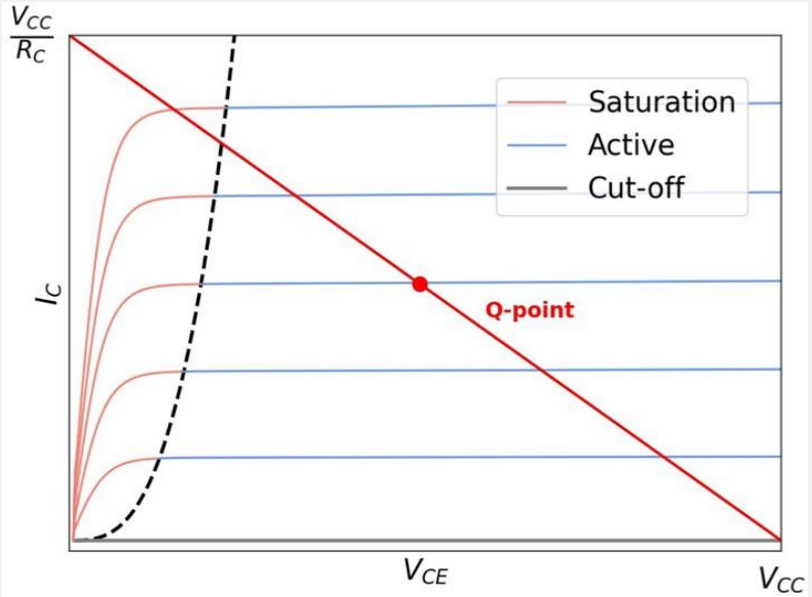
- Find the Q-point from the circuit.

Measure the I_{CQ} & V_{CEQ}


R1
10kΩ
Rc
1kΩ
C
B
E
Vcc
15
Q1
2N2222A
R2
4.7kΩ


I_C
$I_{C(sat)}$
I_{CQ}
Q point
V_{CEQ}
$V_{CE(cutoff)}$
V_{CE}


I_C
2.08 mA
1.236 mA
Q
6.1 V
15 V
V_{CE}


$\frac{V_{CC}}{R_C}$
Saturation
Active
Cut-off
Q-point
I_C
V_{CE}
V_{CC}

Questions

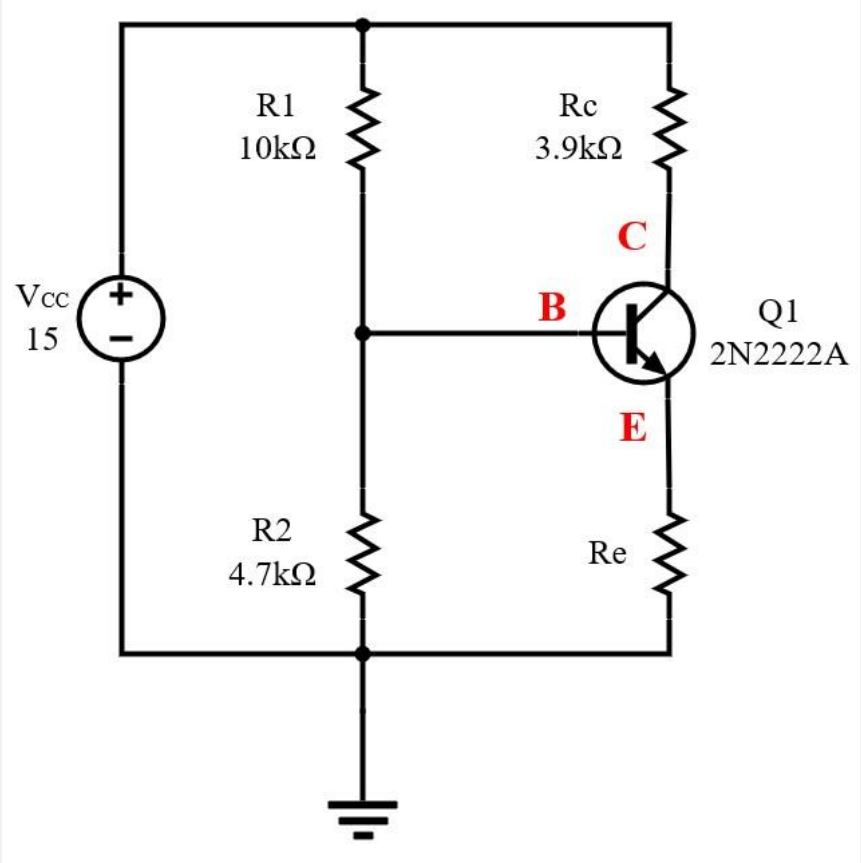
1. Based on the results of Table 2, is the transistor operating in saturation, breakdown or in the active region?

2. Using the original circuit, determine a new value for the collector resistance that will yield a collector voltage of approximately half of the power supply value.

3. If the transistors β value (current gain) decreases from 100 to 50, will the Q-point (I_C, V_{CE}) of the original circuit change significantly? Please explain the reason.

Bonus：Determine the ideal theoretical Q-point.

1. Redesign the circuit to find a new operating point, and record the values of I_C (collector current) and V_{CE} (collector-emitter voltage) at this Q-point.



Common Emitter Amplifier (Voltage Gain)

Obejective

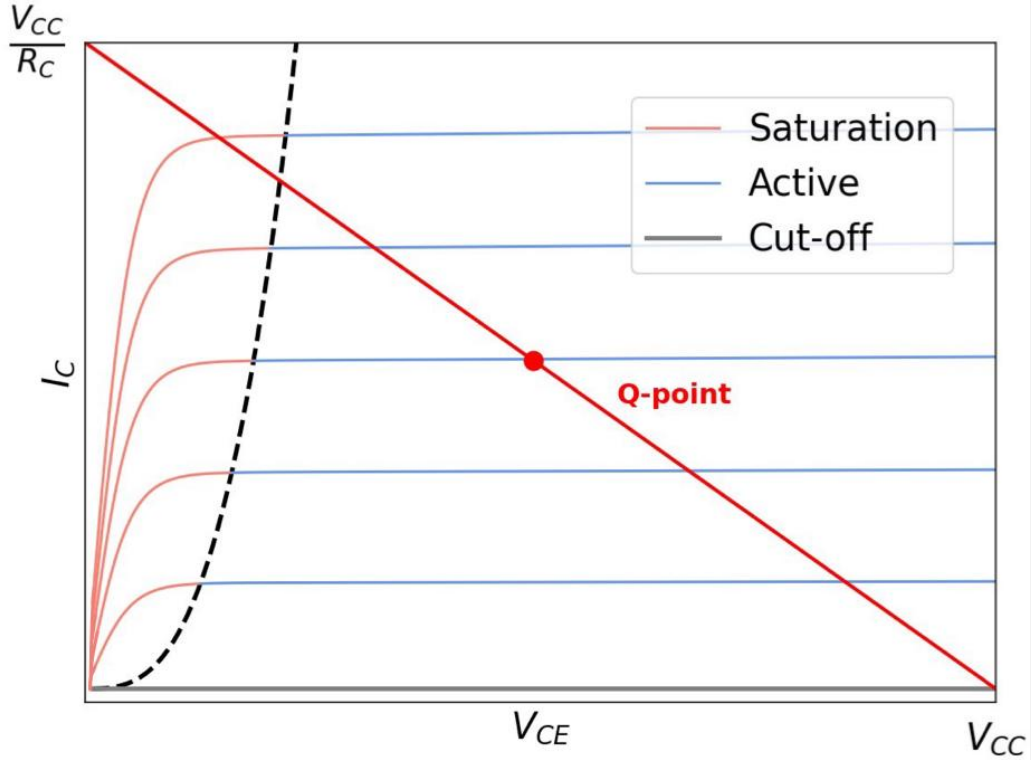
- Understand the influence of transistor β on the voltage gain, input biasing, and signal integrity of a common emitter amplifier.
- Observe input/output phase inversion.
- Examine how circuit parameters (e.g., RC) affect amplifier behavior.
- Develop troubleshooting skills for distortion issues.

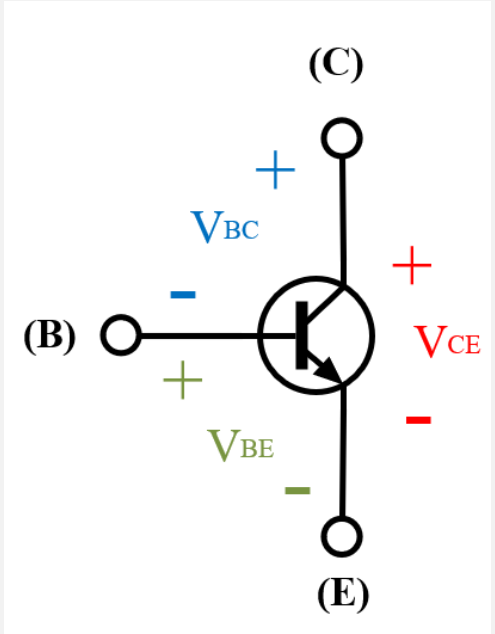
Theory overview

⑩Amplifiers typically operate in the active region.

Mode	BC	BE
Active	$V_{BE} < V_{CE}$ or $V_{CE} < V_{CE(sat.)}$	$V_{BE} \geq 0.7v$

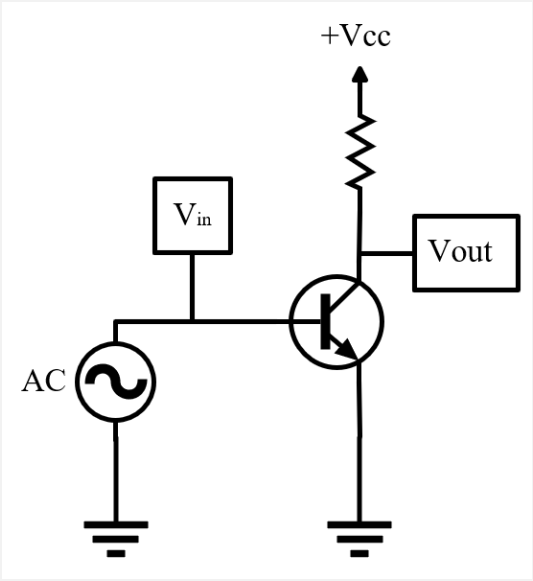
⑩Overdriving the base causes **clipping distortion**.





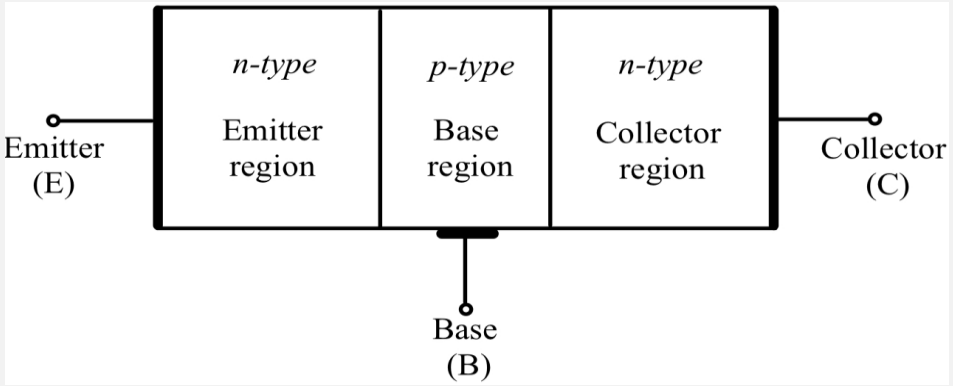
$$*V_{CC} = I_C R_C + V_{CE}$$

$$*I_C = I_S e^{V_{BE}/V_T} = \beta I_B$$



$$*I_E = I_C + I_B = (\beta+1) I_B$$

Theory overview



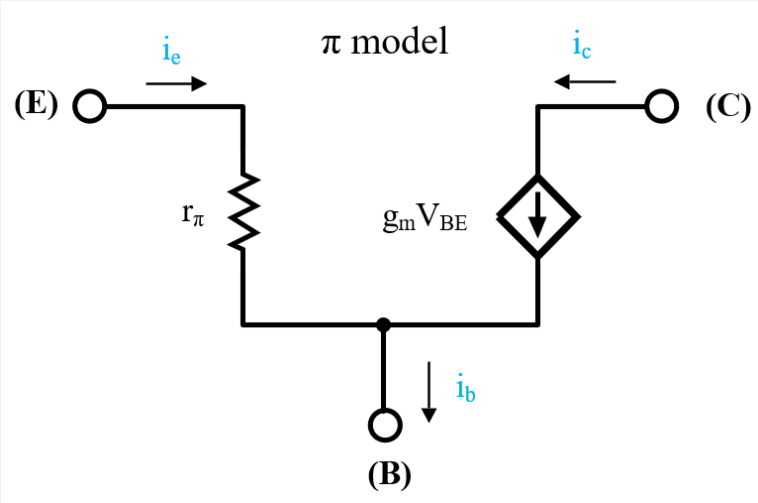
- $g_m = \frac{Ic}{V_T}$

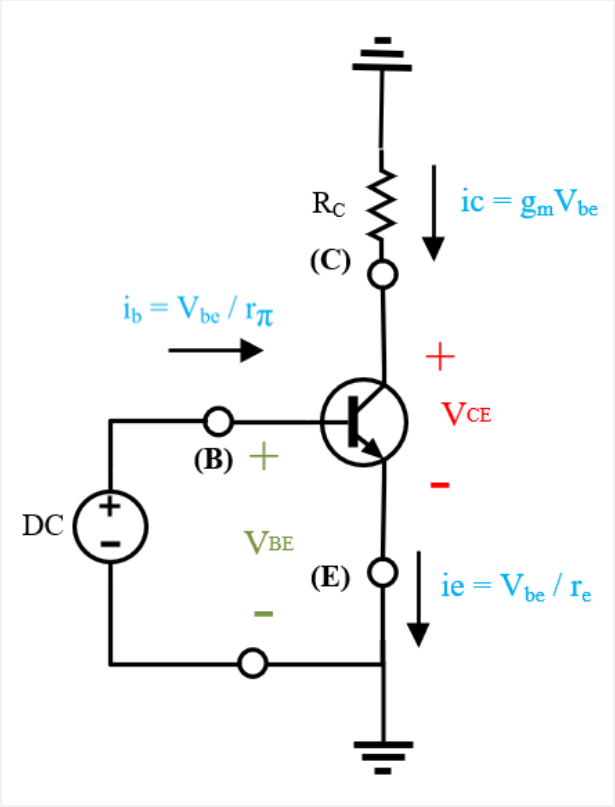
  (VT ≈ 26mV at room temperature 25℃)

- $r_\pi = \frac{v_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$

- $i_C = g_m v_{be}$

- $A_v = \frac{v_{out}}{v_{in}} = \frac{-g_m v_{be} R_C}{v_{be}} = -g_m R_c$





- Larger ***R***c→ higher gain, but may lead to **saturation**.

Component list

Component	Specification	Number
BJT	2N2222A	1
Resistor	100Ω (1/4 W)	2
	1kΩ (1/4 W)	1
	10kΩ (1/4 W)	3
DC Power Supply	GW Instek GPE-3323 (Kaohsiung) UTP-3305-II (Hsinchu)	1
Signal Source (Function Generator)	Keysight EDUX1052G (Kaohsiung) GW Instek AFG-2225 (Hsinchu)	1
Oscilloscope	Keysight EDUX1052G (Kaohsiung) OSD-1102 (Hsinchu)	1
Multimeter	GW Instek GDM-532 (Kaohsiung) DHA DMM-93B+ (Hsinchu)	1
capacitor	1 μF	3

Question

1. What change did you observe in the output waveform after removing the bypass capacitor C3?

2. What are the functions of C1, C2, and C3?

3. How would changing RC1 to a 1kΩ resistor affect the circuit, and why does this happen?

Bonus

Design the values of the capacitors and resistors so that the voltage gain A_V equals 20.

